

Pranjal Nandi

Data Analyst & ML Engineer, Computational Physicist

Based in Germany with working rights, Available Immediately

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Profile

Data analyst and physicist (**PhD, Nanoscience**) with **4+ years** building numerical, statistical and ML workflows for scientific data.

Experienced in **Python, SQL, PyTorch, scikit-learn, Streamlit, PostgreSQL** and **HPC/SLURM** workflows.

Builds ML applications from raw data to **interactive KPI dashboards**, with interests in applied ML, AI systems and scientific data products.

Technical Skills

- Machine Learning**
PyTorch, scikit-learn, forecasting, classification, anomaly detection, model tuning, cross-validation, model evaluation
- Data & Analytics**
Python, SQL, pandas, NumPy, PostgreSQL, data cleaning, feature engineering, KPI dashboards
- AI Applications**
Streamlit, Plotly, ML dashboards, API integration, CSV/JSON export, AI model evaluation
- Software Engineering**
Git & GitHub, CI/CD, pytest, modular Python, documentation, reproducible workflows
- Scientific Computing**
Linux, SLURM, batch computation, Fortran-based ab initio codes, simulation workflows
- Visualisation**
Plotly, matplotlib, spectroscopy, signal/image analysis, FITS workflows
- Languages**
English (C2), German (A1), Spanish (A2)

Achievements

- Selected Young Scientist, 73rd Lindau Nobel Laureate Meeting, 2024
- Catalan Government AGAUR-FI PhD Fellowship
- Full Master's scholarship from Total Energies and Fondation ParisTech

Work Experience

- AI & Machine Learning Engineer** Jan 2026 – Present
Freelancing, Remote
 - Built **ML applications** from raw data to **interactive KPI dashboards** using **Python, SQL, scikit-learn, PyTorch** and **Streamlit**.
 - Developed **forecasting, classification** and **anomaly-detection** workflows with **feature engineering, model evaluation, testing** and **CI/CD practices**.
 - Created **challenging AI-evaluation** questions to evaluate and improve AI models in science, maths, coding and reasoning.
 - Github repository: <https://github.com/pranjal020496?tab=repositories>
- University of Barcelona** Oct 2021 – Nov 2025
PhD Candidate
 - Developed reproducible **Python/SLURM HPC workflows** for simulation automation, **data analysis**, batch execution, result parsing and connection of computational outputs with experimental reference data.
 - Analysed **500+ simulation cases** on convergence and stability of iterative electronic-structure solvers, diagnosing failure modes and establishing **accuracy-versus-runtime trade-offs**.
 - Validated simulated spectra against experimental reference data across **24 pressure-charge configurations** of confined gas materials, identifying spectral signatures and **model-experiment discrepancies**.
 - Built **version-controlled pipelines** for simulation setup, batch execution, result parsing, visualisation and benchmarking across multiple scientific codes.
 - Developed a **simulation framework for defect-dependent HAADF-STEM image analysis** in next-gen battery materials, converting qualitative microscopy contrast into **quantitative defect metrics**.
- University of Vienna** Jun 2024 – Sep 2024
Intern
 - Benchmarked **HPC workloads** and refined spectroscopy models for xenon clusters across **100+ case studies** under varying pressure and charge conditions.
- Swiss Federal Laboratories for Materials, EMPA** Aug 2020 – Dec 2020
Intern
 - Combined experiment, measurement data and root-cause analysis to optimise perovskite solar cells to **21% efficiency**.
- Bharat Aluminium Company Ltd. (Vedanta)** Aug 2018 – Apr 2019
Assistant Manager, Operations and Maintenance
 - Monitored safety-critical systems across **4×300 MW** power-generation units; analysed deviations and coordinated maintenance response.

Education

- University of Barcelona** 2021-2025
PhD in Nanoscience, Faculty of Physics
Thesis: Quantitative Analysis of Functional Materials via Simulation Based Data Frameworks
Spain
- Ecole Polytechnique** 2019-2020
Master's in Renewable Energy Science and Technology, France
- National Power Training Institute** 2014-2018
Bachelors in Power Engineering (interdisciplinary Mechanical and Electrical Engineering)
India